Question ID:181
Topic Name: Mathematics-Section A

Let $f : \mathbb{R} \to \mathbb{R}$ be defined as $f(x) = x - 1$ and $g : \mathbb{R} - \{1, -1\} \to \mathbb{R}$ be defined as $g(x) = \frac{x^2}{x^2 - 1}$.

Question: Then the function fog is :

A one-one but not onto

B onto but not one-one

C both one-one and onto

D neither one-one nor onto

Answer Given By Candidate:**D**

Question ID:182
Topic Name: Mathematics-Section A

If the system of equations
$\alpha x + y + z = 5, x + 2y + 3z = 4, x + 3y + 5z = \beta$
Question: has infinitely many solutions, then the ordered pair (α, β) is equal to :

A $(1, -3)$

B $(-1, 3)$

C $(1, 3)$

D $(-1, -3)$

Answer Given By Candidate:**Not Attempted**

Question ID:183
Topic Name: Mathematics-Section A

Question: If $A = \sum_{n=1}^{\infty} \frac{1}{\left(3+(-1)^n\right)^n}$ and $B = \sum_{n=1}^{\infty} \frac{(-1)^n}{\left(3+(-1)^n\right)^n}$, then $\frac{A}{B}$ is equal to :

A $\frac{11}{9}$

B 1

C $-\frac{11}{9}$

D $-\frac{11}{3}$

Answer Given By Candidate:**Not Attempted**

Question ID:184
Topic Name: Mathematics-Section A

Question: $\lim_{x\to 0} \frac{\cos\left(\sin x\right) - \cos x}{x^4}$ is equal to :

A $\frac{1}{3}$

B $\frac{1}{4}$

C $\frac{1}{6}$

D $\frac{1}{12}$

Answer Given By Candidate:**Not Attempted**

Question ID:185
Topic Name: Mathematics-Section A

Question: Let $f(x) = \min\{1, 1 + x \sin x\}$, $0 \le x \le 2\pi$. If m is the number of points, where f is not differentiable and n is the number of points, where f is not continuous, then the ordered pair (m, n) is equal to

A $(2, 0)$

B $(1, 0)$

C $(1, 1)$

D $(2, 1)$

Answer Given By Candidate:**Not Attempted**

Question ID:186
Topic Name: Mathematics-Section A
Question: Cosider a cuboid of sides $2x$, $4x$ and $5x$ and a closed hemisphere of radius r. If the sum of their surface areas is a constant k, then the ratio $x : r$, for which the sum of their volumes is maximum, is :

A 2:5

B 19:45

C 3:8

D 19:15

Answer Given By Candidate:**Not Attempted**

Question ID:187
Topic Name: Mathematics-Section A
Question: The area of the region bounded by $y^2 = 8x$ and $y^2 = 16(3 - x)$ is equal to:

A $\frac{32}{3}$

B $\frac{40}{3}$

C 16

D 19

Answer Given By Candidate:**Not Attempted**

Question ID:188
Topic Name: Mathematics-Section A
Question: If $\int \frac{1}{x}\sqrt{\frac{1-x}{1+x}}\,dx = g(x) + c$, $g(1) = 0$, then $g\left(\frac{1}{2}\right)$ is equal to :

A $\log_e\left(\frac{\sqrt{3}-1}{\sqrt{3}+1}\right)+\frac{\pi}{3}$

B $\log_e\left(\frac{\sqrt{3}+1}{\sqrt{3}-1}\right)+\frac{\pi}{3}$

C $\log_e\left(\frac{\sqrt{3}+1}{\sqrt{3}-1}\right)-\frac{\pi}{3}$

D

$$\frac{1}{2}\log_e\left(\frac{\sqrt{3}-1}{\sqrt{3}+1}\right)-\frac{\pi}{6}$$

Answer Given By Candidate:**Not Attempted**

Question ID:189
Topic Name: Mathematics-Section A

Question: If $y = y(x)$ is the solution of the differential equation $x\frac{dy}{dx}+2y = x e^x,\ y(1) = 0$ then the local maximum value of the function $z(x) = x^2 y(x) - e^x, x \in \mathbb{R}$ is :

A $1 - e$

B 0

C $\frac{1}{2}$

D $\frac{4}{e}-e$

Answer Given By Candidate:**Not Attempted**

Question ID:1810
Topic Name: Mathematics-Section A

Question: If the solution of the differential equation $\frac{dy}{dx}+e^x\left(x^2-2\right)y=\left(x^2-2x\right)\left(x^2-2\right)e^{2x}$ satisfies $y(0) = 0$, then the value of $y(2)$ is _____

A -1

B 1

C 0

D e

Answer Given By Candidate:**Not Attempted**

Question ID:1811
Topic Name: Mathematics-Section A

Question: If m is the slope of a common tangent to the curves $\frac{x^2}{16}+\frac{y^2}{9}=1$ and $x^2 + y^2 = 12$, then $12\ m^2$ is equal to :

A 6

B 9

C 10

D 12

Answer Given By Candidate:**Not Attempted**

Question ID:1812
Topic Name: Mathematics-Section A

Question: The locus of the mid point of the line segment joining the point (4, 3) and the points on the ellipse $x^2 + 2y^2 = 4$ is an ellipse with eccentricity :

A $\frac{\sqrt{3}}{2}$

B $\frac{1}{2\sqrt{2}}$

C $\frac{1}{\sqrt{2}}$

D $\frac{1}{2}$

Answer Given By Candidate:**Not Attempted**

Question ID:1813
Topic Name: Mathematics-Section A

Question: The normal to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{9} = 1$ at the point $\left(8, 3\sqrt{3}\right)$ on it passes through the point :

A $\left(15, -2\sqrt{3}\right)$

B $\left(9, 2\sqrt{3}\right)$

C $\left(-1, 9\sqrt{3}\right)$

D $\left(-1, 6\sqrt{3}\right)$

Answer Given By Candidate:**Not Attempted**

Question ID:1814
Topic Name: Mathematics-Section A

Question: If the plane $2x + y - 5z = 0$ is rotated about its line of intersection with the plane $3x - y + 4z - 7 = 0$ by an angle of $\frac{\pi}{2}$, then the plane after the rotation passes through the point :

A $(2, -2, 0)$

B $(-2, 2, 0)$

C $(1, 0, 2)$

D $(-1, 0, -2)$

Answer Given By Candidate:**Not Attempted**

Question ID:1815
Topic Name: Mathematics-Section A

Question: If the lines $\vec{r} = \left(\hat{i} - \hat{j} + \hat{k}\right) + \lambda\left(3\hat{j} - \hat{k}\right)$ and $\vec{r} = \left(\alpha\hat{i} - \hat{j}\right) + \mu\left(2\hat{i} - 3\hat{k}\right)$ are co-planar, then the distance of the plane containing these two lines from the point $(\alpha, 0, 0)$ is:

A $\frac{2}{9}$

B $\frac{2}{11}$

C $\frac{4}{11}$

D 2

Answer Given By Candidate:**Not Attempted**

Question ID:1816
Topic Name: Mathematics-Section A

Question: Let $\vec{a} = \hat{i} + \hat{j} + 2\hat{k}$, $\vec{b} = 2\hat{i} - 3\hat{j} + \hat{k}$ and $\vec{c} = \hat{i} - \hat{j} + \hat{k}$ be three given vectors. Let $\vec{v}$ be a vector in the plane of $\vec{a}$ and $\vec{b}$ whose projection on $\vec{c}$ is $\frac{2}{\sqrt{3}}$. If $\vec{v} \cdot \hat{j} = 7$, then $\vec{v} \cdot \left(\hat{i} + \hat{k}\right)$ is equal to :

A 6

B 7

C 8

D 9

Answer Given By Candidate:**Not Attempted**

Question ID:1817
Topic Name: Mathematics-Section A
Question: The mean and standard deviation of 50 observations are 15 and 2 respectively. It was found that one incorrect observation was taken such that the sum of correct and incorrect observations is 70. If the correct mean is 16, then the correct variance is equal to :

A 10

B 36

C 43

D 60

Answer Given By Candidate:**Not Attempted**

Question ID:1818
Topic Name: Mathematics-Section A
Question: $16\sin(20^\circ)\sin(40^\circ)\sin(80^\circ)$ is equal to :

A $\sqrt{3}$

B $2\sqrt{3}$

C 3

D $4\sqrt{3}$

Answer Given By Candidate:**Not Attempted**

Question ID:1819
Topic Name: Mathematics-Section A
Question: If the inverse trignometric functions take principal values, then $\cos^{-1}\left(\frac{3}{10}\cos\left(\tan^{-1}\left(\frac{4}{3}\right)\right)+\frac{2}{5}\sin\left(\tan^{-1}\left(\frac{4}{3}\right)\right)\right)$ is equal to :

A 0

B $\frac{\pi}{4}$

C $\frac{\pi}{3}$

D

$\frac{\pi}{6}$

Answer Given By Candidate:**Not Attempted**

Question ID:1820
Topic Name: Mathematics-Section A

Question: Let $r \in \{p, q, \sim p, \sim q\}$ be such that the logical statement $r \vee (\sim p) \Rightarrow (p \wedge q) \vee r$ is a tautology. Then r is equal to :

A p

B q

C $\sim p$

D $\sim q$

Answer Given By Candidate:**Not Attempted**

Question ID:1821
Topic Name: Mathematics-Section B

Question: Let $f: \mathbb{R} \to \mathbb{R}$ satisfy $f(x+y) = 2^x f(y) + 4^y f(x), \forall x, y \in \mathbb{R}$. If $f(2) = 3$, then $14 \cdot \frac{f'(4)}{f'(2)}$ is equal to _____.

Answer Given By Candidate:**Not Attempted**

Question ID:1822
Topic Name: Mathematics-Section B

Question: Let p and q be two real numbers such that $p + q = 3$ and $p^4 + q^4 = 369$. Then $\left(\frac{1}{p} + \frac{1}{q}\right)^{-2}$ is equal to _____.

Answer Given By Candidate:**Not Attempted**

Question ID:1823
Topic Name: Mathematics-Section B

Question: If $z^2 + z + 1 = 0, z \in \mathbb{C}$, then $\left| \sum_{n=1}^{15} \left(z^n + (-1)^n \frac{1}{z^n} \right)^2 \right|$ is equal to _____.

Answer Given By Candidate:**Not Attempted**

Question ID:1824
Topic Name: Mathematics-Section B

Let $X=\begin{bmatrix}0&1&0\\0&0&1\\0&0&0\end{bmatrix}$, $Y=\alpha I+\beta X+\gamma X^2$ and

$Z=\alpha^2 I-\alpha\beta X+(\beta^2-\alpha\gamma)X^2$, $\alpha,\beta,\gamma\in\mathbb{R}$. If $Y^{-1}=\begin{bmatrix}1/5&-2/5&1/5\\0&1/5&-2/5\\0&0&1/5\end{bmatrix}$, then

Question: $(\alpha-\beta+\gamma)^2$ is equal to _____.

Answer Given By Candidate:**Not Attempted**

Question ID:1825
Topic Name: Mathematics-Section B

The total number of 3-digit numbers, whose greatest common divisor with 36 is 2,

Question: is _____.

Answer Given By Candidate:**Not Attempted**

Question ID:1826
Topic Name: Mathematics-Section B

If $\left({}^{40}C_0\right)+\left({}^{41}C_1\right)+\left({}^{42}C_2\right)+....+\left({}^{60}C_{20}\right)=\frac{m}{n}\,{}^{60}C_{20}$ m and n are coprime, then

Question: $m+n$ is equal to _____.

Answer Given By Candidate:**Not Attempted**

Question ID:1827
Topic Name: Mathematics-Section B

If $a_1\ (>0)$, a_2, a_3, a_4, a_5 are in a G.P., $a_2+a_4=2a_3+1$ and $3a_2+a_3=2a_4$, then

Question: $a_2+a_4+2a_5$ is equal to _____.

Answer Given By Candidate:**Not Attempted**

Question ID:1828
Topic Name: Mathematics-Section B

Question: The integral $\frac{24}{\pi}\int_0^{\sqrt{2}}\frac{\left(2-x^2\right)dx}{\left(2+x^2\right)\sqrt{4+x^4}}$ is equal to _____.

Answer Given By Candidate:**Not Attempted**

Question ID:1829
Topic Name: Mathematics-Section B

Question: Let a line L_1 be tangent to the hyperbola $\frac{x^2}{16}-\frac{y^2}{4}=1$ and let L_2 be the line passing through the origin and perpendicular to L_1. If the locus of the point of intersection of L_1 and L_2 is $(x^2+y^2)^2=\alpha x^2+\beta y^2$, then $\alpha+\beta$ is equal to _____.

Answer Given By Candidate:**Not Attempted**

Question ID:1830
Topic Name: Mathematics-Section B
Question: If the probability that a randomly chosen 6-digit number formed by using digits 1 and 8 only is a multiple of 21 is p, then $96p$ is equal to _____.

Answer Given By Candidate:**Not Attempted**

Question ID:1831
Topic Name: Physics-Section A
Question: The dimension of mutual inductance is :

A $[ML^2T^{-2}A^{-1}]$

B $[ML^2T^{-3}A^{-1}]$

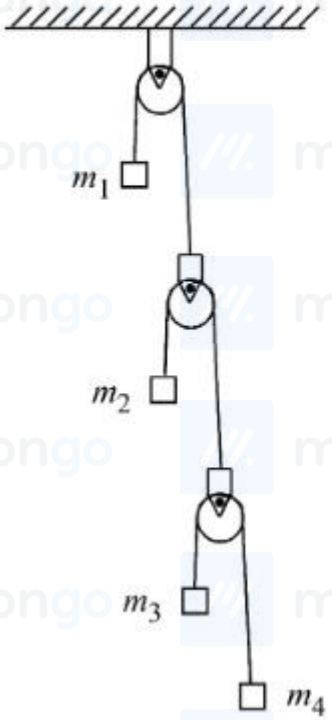
C $[ML^2T^{-2}A^{-2}]$

D $[ML^2T^{-3}A^{-2}]$

Answer Given By Candidate:**Not Attempted**

Question ID:1832
Topic Name: Physics-Section A

In the arrangement shown in figure a_1, a_2, a_3 and a_4 are the accelerations of masses m_1, m_2, m_3 and m_4 respectively. Which of the following relation is true for this arrangement ?



Question:

A $4a_1 + 2a_2 + a_3 + a_4 = 0$

B $a_1 + 4a_2 + 3a_3 + a_4 = 0$

C $a_1 + 4a_2 + 3a_3 + 2a_4 = 0$

D $2a_1 + 2a_2 + 3a_3 + a_4 = 0$

Answer Given By Candidate:**A**

Question ID:1833
Topic Name: Physics-Section A

Arrange the four graphs in descending order of total work done; where W_1, W_2, W_3 and W_4 are the work done corresponding to figure a, b, c and d respectively.

Figure-a

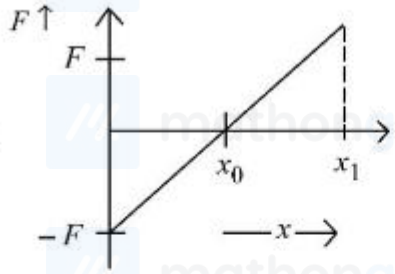


Figure-b

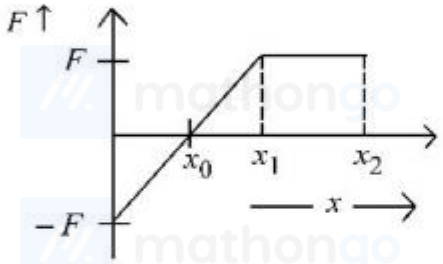


Figure-c

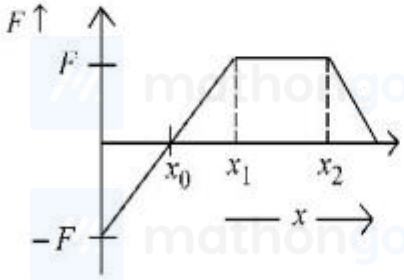
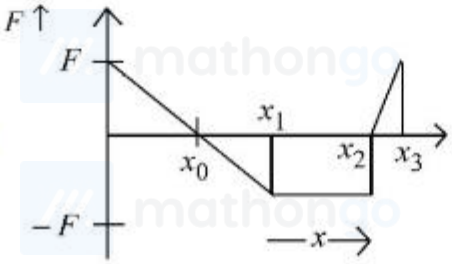


Figure-d



Question:

A $W_3 > W_2 > W_1 > W_4$

B $W_3 > W_2 > W_4 > W_1$

C $W_2 > W_3 > W_4 > W_1$

D $W_2 > W_3 > W_1 > W_4$

Answer Given By Candidate:**C**

Question ID:1834
Topic Name: Physics-Section A

Question: A solid spherical ball is rolling on a frictionless horizontal plane surface about its axis of symmetry. The ratio of rotational kinetic energy of the ball to its total kinetic energy is -

A $\frac{2}{5}$

B $\frac{2}{7}$

C $\frac{1}{5}$

D $\frac{7}{10}$

Answer Given By Candidate:**Not Attempted**

Question ID:1835
Topic Name: Physics-Section A

Question: Given below are two statements: One is labelled as Assertion A and the other is labelled as Reason R.

Assertion A: If we move from poles to equator, the direction of acceleration due to gravity of earth always points towards the center of earth without any variation in its magnitude.
Reason R: At equator, the direction of acceleration due to the gravity is towards the center of earth.

In the light of above statements, choose the correct answer from the options given below

A Both A and R are true and R is the correct explanation of A.

B Both A and R are true but R is NOT the correct explanation of A.

C A is true but R is false

D A is false but R is true

Answer Given By Candidate:**B**

Question ID:1836
Topic Name: Physics-Section A

Question: If p is the density and η is coefficient of viscosity of fluid which flows with a speed v in the pipe of diameter d, the correct formula for Reynolds number R_e is :

A $R_e = \frac{\eta d}{\rho v}$

B $R_e = \frac{\rho\nu}{\eta d}$

C $R_e = \frac{\rho\nu d}{\eta}$

D $R_e = \frac{\eta}{\rho\nu d}$

Answer Given By Candidate:**Not Attempted**

Question ID:1837
Topic Name: Physics-Section A

A flask contains argon and oxygen in the ratio of 3:2 in mass and the mixture is kept at 27°C. The ratio of their average kinetic energy per molecule respectively

Question: will be :

A 3:2

B 9:4

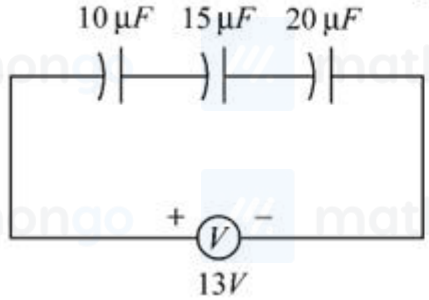
C 2:3

D 1:1

Answer Given By Candidate:**C**

Question ID:1838
Topic Name: Physics-Section A

The charge on capacitor of capacitance $15\mu F$ in the figure given below is :



Question:

A $60\mu c$

B $130\mu c$

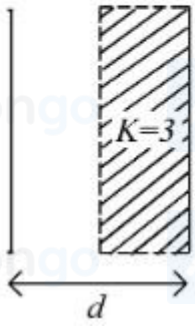
C $260\mu c$

D $585\mu c$

Answer Given By Candidate:**A**

Question ID:1839
Topic Name: Physics-Section A

Question: A parallel plate capacitor with plate area A and plate separation d=2 m has a capacitance of 4 μF. The new capacitance of the system if half of the space between them is filled with a dielectric material of dielectric constant K=3 (as shown in figure) will be :



A $2\mu F$

B $32\mu F$

C $6\mu F$

D $8\mu F$

Answer Given By Candidate:**Not Attempted**

Question ID:1840
Topic Name: Physics-Section A

Question: Sixty four conducting drops each of radius 0.02 m and each carrying a charge of 5 μC are combined to form a bigger drop. The ratio of surface density of bigger drop to the smaller drop will be :

A 1:4

B 4:1

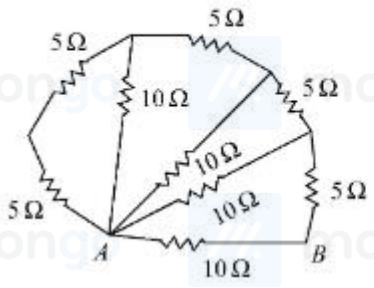
C 1:8

D 8:1

Answer Given By Candidate:**Not Attempted**

Question ID:1841
Topic Name: Physics-Section A

The equivalent resistance between points A and B in the given network is :



Question:

A 65Ω

B 20Ω

C 5Ω

D 2Ω

Answer Given By Candidate:**C**

Question ID:1842
Topic Name: Physics-Section A

A bar magnet having a magnetic moment of 2.0×10^5 JT^{-1}, is placed along the direction of uniform magnetic field of magnitude $B=14 \times 10^{-5}\ T$. The work done in rotating the magnet slowly through 60° from the direction of field is :

Question:

A 14 J

B 8.4 J

C 4 J

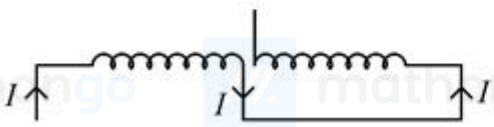
D 1.4 J

Answer Given By Candidate:**Not Attempted**

Question ID:1843
Topic Name: Physics-Section A

Two coils of self inductance L_1 and L_2 are connected in series combination having mutual inductance of the coils as M. The equivalent self inductance of the combination will be :

Question:



A $\frac{1}{L_1}+\frac{1}{L_2}+\frac{1}{M}$

B $L_1 + L_2 + M$

C $L_1 + L_2 + 2M$

D $L_1 + L_2 - 2M$

Answer Given By Candidate:**Not Attempted**

Question ID:1844
Topic Name: Physics-Section A
Question: A metallic conductor of length 1m rotates in a vertical plane parallel to east-west direction about one of its end with angular velocity 5 $rad\ s^{-1}$. If the horizontal component of earth's magnetic field is $0.2 \times 10^{-4}\ T$, then emf induced between the two ends of the conductor is :

A $5\mu V$

B $50\mu V$

C $5mV$

D $50mV$

Answer Given By Candidate:**Not Attempted**

Question ID:1845
Topic Name: Physics-Section A
Question: Which is the correct ascending order of wavelengths ?

A $\lambda_{visible} < \lambda_{X\text{-}ray} < \lambda_{gamma\text{-}ray} < \lambda_{microwave}$

B $\lambda_{gamma\text{-}ray} < \lambda_{X\text{-}ray} < \lambda_{visible} < \lambda_{microwave}$

C $\lambda_{X\text{-}ray} < \lambda_{gamma\text{-}ray} < \lambda_{visible} < \lambda_{microwave}$

D $\lambda_{microwave} < \lambda_{visible} < \lambda_{gamma\text{-}ray} < \lambda_{X\text{-}ray}$

Answer Given By Candidate:**B**

Question ID:1846
Topic Name: Physics-Section A
Question: For a specific wavelength 670 nm of light coming from a galaxy moving with velocity v, the observed wavelength is 670.7 nm. The value of v is :

A $3 \times 10^8\ ms^{-1}$

B $3 \times 10^{10}\ ms^{-1}$

C $3.13 \times 10^5\ ms^{-1}$

D $4.48 \times 10^5\ ms^{-1}$

Answer Given By Candidate:**Not Attempted**

Question ID:1847
Topic Name: Physics-Section A

Question: A metal surface is illuminated by a radiation of wavelength 4500 Å. The ejected photo-electron enters a constant magnetic field of 2 mT making an angle of 90°with the magnetic field. If it starts revolving in a circular path of radius 2 mm, the work function of the metal is approximately :

A 1.36 eV

B 1.69 eV

C 2.78 eV

D 2.23 eV

Answer Given By Candidate:**Not Attempted**

Question ID:1848
Topic Name: Physics-Section A

Question: A radioactive nucleus can decay by two different processes. Half-life for the first process is 3.0 hours while it is 4.5 hours for the second process. The effective half-life of the nucleus will be :

A 3.75 hours

B 0.56 hours

C 0.26 hours

D 1.80 hours

Answer Given By Candidate:**D**

Question ID:1849
Topic Name: Physics-Section A

Question: The positive feedback is required by an amplifier to act an oscillator. The feedback here means :

A External input is necessary to sustain ac signal in output.

B A portion of the output power is returned back to the input.

C Feedback can be achieved by LR network.

D The base-collector junction must be forward biased.

Answer Given By Candidate:**Not Attempted**

Question ID:1850
Topic Name: Physics-Section A

Question: A sinusoidal wave $y(t) = 40\sin(10 \times 10^6\ \pi t)$ is amplitude modulated by another sinusoidal wave $x(t) = 20\sin(1000\pi t)$. The amplitude of minimum frequency component of modulated signal is :

A 0.5

B 0.25

C 20

D 10

Answer Given By Candidate:**Not Attempted**

Question ID:1851
Topic Name: Physics-Section B

Question: A ball is projected vertically upward with an initial velocity of $50\ ms^{-1}$ at $t = 0s$. At $t = 2s$, another ball is projected vertically upward with same velocity. At $t =$ _____s, second ball will meet the first ball ($g = 10\ ms^{-2}$).

Answer Given By Candidate:**6**

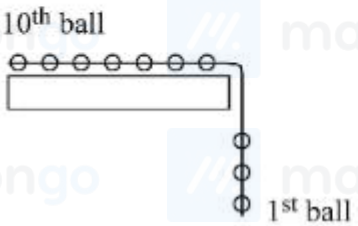
Question ID:1852
Topic Name: Physics-Section B

Question: A batsman hits back a ball of mass 0.4 kg straight in the direction of the bowler without changing its initial speed of $15\ ms^{-1}$. The impulse imparted to the ball is _____ Ns.

Answer Given By Candidate:**12**

Question ID:1853
Topic Name: Physics-Section B

Question: A system to 10 balls each of mass 2 kg are connected via massless and unstretchable string. The system is allowed to slip over the edge of a smooth table as shown in figure. Tension on the string between the 7th and 8th ball is _____N when 6th ball just leaves the table.



Answer Given By Candidate:**120**

Question ID:1854
Topic Name: Physics-Section B

Question: A geyser heats water flowing at a rate of 2.0 kg per minute from 30°C to 70°C. If geyser operates on a gas burner, the rate of combustion of fuel will be _____ $g\ min^{-1}$
[Heat of combustion = $8 \times 10^3\ Jg^{-1}$, Specific heat of water = $4.2\ Jg^{-1}\ °C^{-1}$]

Answer Given By Candidate:**Not Attempted**

Question ID:1855
Topic Name: Physics-Section B

Question: A heat engine operates with the cold reservoir at temperature 324 K. The minimum temperature of the hot reservoir, if the heat engine takes 300 J heat from the hot reservoir and delivers 180 J heat to the cold reservoir per cycle, is _____ K.

Answer Given By Candidate:**Not Attempted**

Question ID:1856
Topic Name: Physics-Section B

Question: A set of 20 tuning forks is arranged in a series of increasing frequencies. If each fork gives 4 beats with respect to the preceding fork and the frequency of the last fork is twice the frequency of the first, then the frequency of last fork is _____ Hz.

Answer Given By Candidate:**Not Attempted**

Question ID:1857
Topic Name: Physics-Section B

Question: Two 10 cm long, straight wires, each carrying a current of 5A are kept parallel to each other. If each wire experienced a force of $10^{-5}\ N$, then separation between the wires is _____ cm.

Answer Given By Candidate:**Not Attempted**

Question ID:1858
Topic Name: Physics-Section B

Question: A small bulb is placed at the bottom of a tank containing water to a depth of $\sqrt{7}\ m$. The refractive index of water is $\frac{4}{3}$. The area of the surface of water through which light from the bulb can emerge out is $x\pi\ m^2$. The value of x is _____.

Answer Given By Candidate:**Not Attempted**

Question ID:1859
Topic Name: Physics-Section B

Question: A travelling microscope is used to determine the refractive index of a glass slab. If 40 divisions are there in 1 cm on main scale and 50 Vernier scale divisions are equal to 49 main scale divisions, then least count of the travelling microscope is _____ $\times 10^{-6}$ m.

Answer Given By Candidate:**Not Attempted**

Question ID:1860
Topic Name: Physics-Section B

Question: The stopping potential for photoelectrons emitted from a surface illuminated by light of wavelength 6630 Å is 0.42 V. If the threshold frequency is $x \times 10^{13}$ $/s$, where x is _____ (nearest integer).
(Given, speed light = 3×10^{8} m/s ,Planck's constant = 6.63×10^{-34} Js)

Answer Given By Candidate:**Not Attempted**

Question ID:1861
Topic Name: Chemistry-Section A

Question: The number of radial and angular nodes in 4d orbital are, respectively

A 1 and 2

B 3 and 2

C 1 and 0

D 2 and 1

Answer Given By Candidate:**D**

Question ID:1862
Topic Name: Chemistry-Section A

Match **List I** with **List II**.

List I Enzyme	List II Conversion of
A. Invertase	I. Starch into maltose
B. Zymase	II. Maltose into glucose
C. Diastase	III. Glucose into ethanol
D. Maltase	IV. Cane sugar into glucose

Question: Choose the most appropriate answer from the options given below:

A A-III, B-IV, C-II, D-I

B A-III, B-II, C-I, D-IV

C A-IV, B-III, C-I, D-II

D A-IV, B-II, C-III, D-I

Answer Given By Candidate:**Not Attempted**

Question ID:1863
Topic Name: Chemistry-Section A
Question: Which of the following elements is considered as a metalloid?

A Sc

B Pb

C Bi

D Te

Answer Given By Candidate:**B**

Question ID:1864
Topic Name: Chemistry-Section A
Question: The role of depressants in 'Froth Floation method' is to

A selectively prevent one component of the ore from coming to the froth.

B reduce the consumption of oil for froth formation.

C stabilize the froth.

D enhance non-wettability of the mineral particles.

Answer Given By Candidate:**A**

Question ID:1865
Topic Name: Chemistry-Section A
Question: Boiling of hard water is helpful in removing the temporary hardness by converting calcium hydrogen carbonate and magnesium hydrogen carbonate to

A $CaCO_3$ and $Mg(OH)_2$

B $CaCO_3$ and $MgCO_3$

C $Ca(OH)_2$ and $MgCO_3$

D $Ca(OH)_2$ and $Mg(OH)_2$

Answer Given By Candidate:**B**

Question ID:1866
Topic Name: Chemistry-Section A
Question: s-block element which **cannot** be qualitatively confirmed by the flame test is

A Li

B Na

C Rb

D Be

Answer Given By Candidate:**D**

Question ID:1867
Topic Name: Chemistry-Section A
Question: The oxide which contains an odd electron at the nitrogen atom is

A N_2O

B NO_2

C N_2O_3

D N_2O_5

Answer Given By Candidate:**B**

Question ID:1868
Topic Name: Chemistry-Section A
Question: Which one of the following is an example of disproportionation reaction ?

A

$3MnO_4^{2-} + 4H^+ \rightarrow 2MnO_4^- + MnO_2 + 2H_2O$

B $MnO_4^- + 4H^+ + 4e^- \rightarrow MnO_2 + 2H_2O$

C $10I^- + 2MnO_4^- + 16H^+ \rightarrow 2Mn^{2+} + 8H_2O + 5I_2$

D $8MnO_4^- + 3S_2O_3^{2-} + H_2O \rightarrow 8MnO_2 + 6SO_4^{2-} + 2OH^-$

Answer Given By Candidate:**Not Attempted**

Question ID:1869
Topic Name: Chemistry-Section A
Question: The most common oxidation state of Lanthanoid elements is +3. Which of the following is likely to deviate easily from +3 oxidation state?

A Ce (At. No. 58)

B La (At. No. 57)

C Lu (At. No. 71)

D Gd (At. No. 64)

Answer Given By Candidate:**Not Attempted**

Question ID:1870
Topic Name: Chemistry-Section A
Question: The measured BOD values for four different water samples (A-D) are as follows: A = 3 ppm; B=18 ppm; C=21 ppm; D=4 ppm. The water samples which can be called as highly polluted with organic wastes, are

A A and B

B A and D

C B and C

D B and D

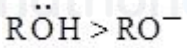
Answer Given By Candidate:**D**

Question ID:1871
Topic Name: Chemistry-Section A
Question: The correct order of nucleophilicity is

A $F^- > OH^-$

B $H_2\ddot{O} > OH^-$

C



D $NH_2^- > NH_3$

Answer Given By Candidate:**Not Attempted**

Question ID:1872
Topic Name: Chemistry-Section A
Question: Oxidation of toluene to benzaldehyde can be easily carried out with which of the following reagents?

A CrO_3/acetic acid, H_3O^+

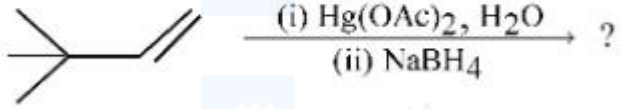
B CrO_3/acetic anhydride, H_3O^+

C $KMnO_4$/HCl, H_3O^+

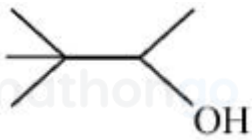
D CO/HCl, anhydrous $AlCl_3$

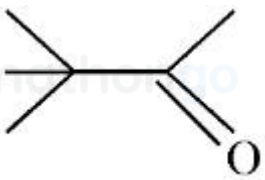
Answer Given By Candidate:**C**

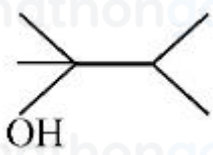
Question ID:1873
Topic Name: Chemistry-Section A
Question: The major product in the following reaction



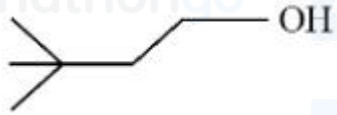
is

A


B


C


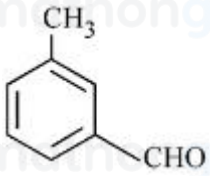
D

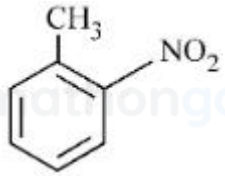


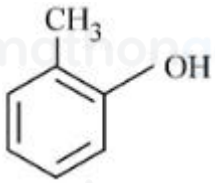
Answer Given By Candidate:**D**

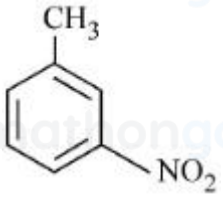
Question ID:1874
Topic Name: Chemistry-Section A

Question: Halogenation of which one of the following will yield m-substituted product with respect to methyl group as a major product?

A 

B 

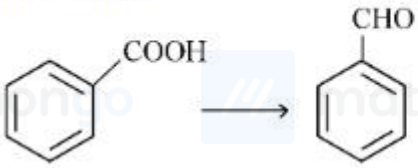
C 

D 

Answer Given By Candidate:**D**

Question ID:1875
Topic Name: Chemistry-Section A

Question: The reagent, from the following, which converts benzoic acid to benzaldehyde in one step is



A $LiAlH_4$

B $KMnO_4$

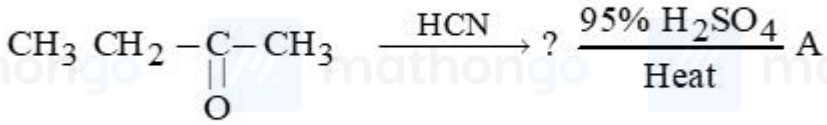
C MnO

D $NaBH_4$

Answer Given By Candidate:**B**

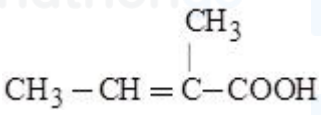
Question ID:1876
Topic Name: Chemistry-Section A

The final product 'A' in the following reaction sequence

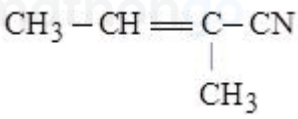


Question:is

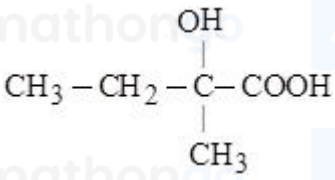
A



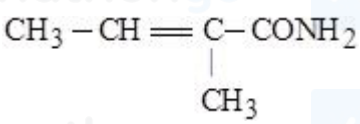
B



C



D



Answer Given By Candidate:**Not Attempted**

Question ID:1877
Topic Name: Chemistry-Section A
Question: Which statement is NOT correct for p-toluenesulphonyl chloride?

A It is known as Hinsberg's reagent.

B It is used to distinguish primary and secondary amines.

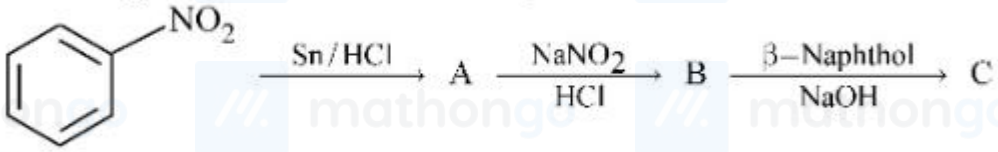
C On treatment with secondary amine, it leads to a product, that is soluble in alkali.

D It doesn't react with tertiary amines.

Answer Given By Candidate:**Not Attempted**

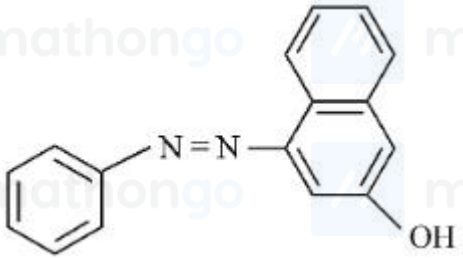
Question ID:1878
Topic Name: Chemistry-Section A

The final product 'C' in the following series of reactions

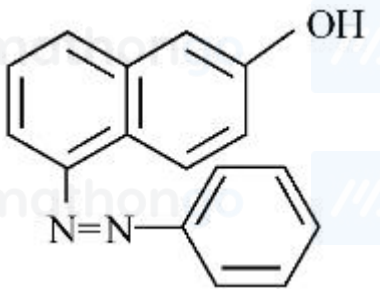


Question: is

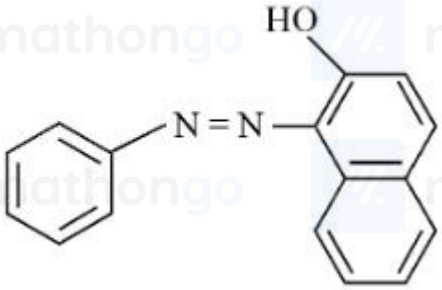
A



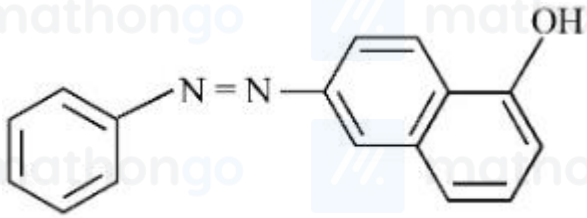
B



C



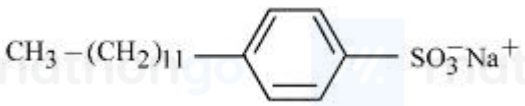
D



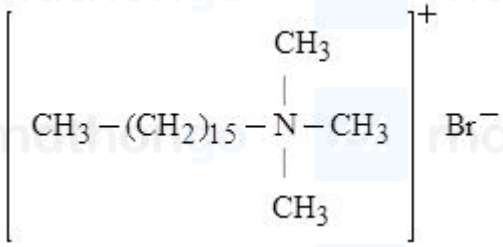
Answer Given By Candidate: **Not Attempted**

Question ID: 1879
Topic Name: Chemistry-Section A
Question: Which of the following is NOT an example of synthetic detergent?

A



B $CH_3-(CH_2)_{16}-COO^- Na^+$

C 

D $CH_3(CH_2)_{16}COO(CH_2\,CH_2O)_nCH_2CH_2OH$

Answer Given By Candidate:**D**

Question ID:1880
Topic Name: Chemistry-Section A
Question: Which one of the following is a water soluble vitamin, that is not excreted easily?

A Vitamin B_2

B Vitamin B_1

C Vitamin B_6

D Vitamin B_{12}

Answer Given By Candidate:**Not Attempted**

Question ID:1881
Topic Name: Chemistry-Section B

CNG is an important transportation fuel. When 100 g CNG is mixed with 208 g oxygen in vehicles, it leads to the formation of CO_2 and H_2O and produces large quantity of heat during this combustion ,then the amount of carbon dioxide, produced in grams is ____. [nearest integer]

Question: [Assume CNG to be methane]

Answer Given By Candidate:**Not Attempted**

Question ID:1882
Topic Name: Chemistry-Section B

In a solid AB, A atoms are in ccp arrangement and B atoms occupy all the octahedral sites. If two atoms from the opposite faces are removed, then the resultant stoichiometry of the compound is A_xB_y. The value of x is ____. [nearest integer]

Question:

Answer Given By Candidate:**Not Attempted**

Question ID:1883
Topic Name: Chemistry-Section B

Question: Amongst SF_4, XeF_4, CF_4 and H_2O, the number of species with two lone pairs of electrons is ____.

Answer Given By Candidate:**2**

Question ID:1884
Topic Name: Chemistry-Section B

Question: A fish swimming in water body when taken out from the water body is covered with a film of water of weight 36 g. When it is subjected to cooking at 100 °C, then the internal energy for vaporization in kJ mol^{-1} is _____. [nearest integer]

[Assume steam to be an ideal gas. Given $\Delta_{vap}H^{\ominus}$ for water at 373 K and 1 bar is 41.1 kJ mol^{-1} ; R = 8.31 J $K^{-1}mol^{-1}$]

Answer Given By Candidate:**Not Attempted**

Question ID:1885
Topic Name: Chemistry-Section B

Question: The osmotic pressure exerted by a solution prepared by dissolving 2.0 g of protein of molar mass 60 kg mol^{-1} in 200 mL of water at 27°C is _____ Pa.[integer value]

(use R = 0.083 L bar mol^{-1} K^{-1})

Answer Given By Candidate:**42**

Question ID:1886
Topic Name: Chemistry-Section B

Question: 40% of HI undergoes decomposition to H_2 and I_2 at 300 K. $\Delta G^{\ominus}$ for this decompostion reaction at one atmopsphere pressure is _____ J mol^{-1}.[nearest integer]

(Use R = 8.31 J K^{-1} mol^{-1}; log 2 = 0.3010, ln 10 =2.3, log 3 = 0.477)

Answer Given By Candidate:**Not Attempted**

Question ID:1887
Topic Name: Chemistry-Section B

Question: $Cu(s) + Sn^{2+}(0.001M) \rightarrow Cu^{2+}(0.01M) + Sn(s)$

The Gibbs free energy change for the above reaction at 298 K is $x \times 10^{-1}$ kJ mol^{-1}. The value of x is _____. [nearest integer]

$\left[\text{Given : } E^{\ominus}_{Cu^{2+}/Cu} = 0.34\,V\ ;\ E^{\ominus}_{Sn^{2+}/Sn} = -0.14\,V\ ;\ F = 96500\ C\ mol^{-1}\right]$

Answer Given By Candidate:**Not Attempted**

Question ID:1888
Topic Name: Chemistry-Section B

Question: Catalyst A reduces the activation energy for a reaction by 10 kJ mol^{-1} at 300 K. The ratio of rate constants, $\frac{k_{T,\,Catalysed}}{k_{T,\,Uncatalysed}}$ is e^{x}. The value of x is ___. [nearest integer]

[Assume that the pre-exponential factor is same in both the cases. Given R = 8.31 J $K^{-1}mol^{-1}$]

Answer Given By Candidate:**Not Attempted**

Question ID:1889
Topic Name: Chemistry-Section B

Question: Reaction of $[Co(H_2O)_6]^{2+}$ with excess ammonia and in the presence of oxygen results into a diamagnetic product. Number of electrons present in t_{2g}-orbitals of the product is _____.

Answer Given By Candidate:**00**

Question ID:1890
Topic Name: Chemistry-Section B

Question: The moles of methane required to produce 81 g of water after complete combustion is _____ $\times 10^{-2}$ mol. [nearest integer]

Answer Given By Candidate:**18**